Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/03/2006
PI Name:	Smith, Fredrick D B.S.		
Project Title:	ARTEF		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	frederick.d.smith1@jsc.nasa.gov	**Fax:**	FY 281 483-5060
PI Organization Type:	NASA CENTER	**Phone:**	281 483-9232
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Mail Code EC3		
PI Address 2:	Building 7, Room 222		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	10/01/2001	**End Date:**	10/01/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: changed end date to 10/1/2007, from 9/01/2010, because of project reporting no longer required (10/07 info fr program mgr)--10/8/08		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Advanced Life Support (ALS) Air Revitalization Element Technology Evaluation Facility (ARTEF) will facilitate the operation and testing of individual components, as well as the fully integrated performance of Air Revitalization technology. As an integral part of a complete Environmental Control Life Support (ECLS) System, the lab will also facilitate the integration of the Air Revitalization System (ARS) Element with other complementary ALS Elements. The ARTEF will typically be used to evaluate hardware with a Technology Readiness Level (TRL) between 3 and 5. The ARTEF will accommodate the bench level operation and testing of individual prototype and Engineering Development Unit (EDU) components, the long term testing of prototype and EDU components, the end-to-end operation and testing of integrated ARS subsystems, the fully integrated, long term testing of ARS subsystems, and facilitate the integration of the ARS with other complementary ECLS subsystems. The ARTEF will also provide capability for simulated and actual human metabolic loads.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This area is a facility capability however the technologies that are evaluated in this facility will have potential benefit to Earth in the air purification arena.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)